

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461057.763 +/- 0.012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.091 +/- 0.039
Transit depth (Rp/Rs)^2	0.82 +/- 0.71 [%]
Orbital Inclination (inc)	85.84 +/- 2.66
Airmass coefficient 1 (a1)	0.873 +/- 0.005
Airmass coefficient 2 (a2)	0.099 +/- 0.088
Scatter in the residuals of the lightcurve fit is	0.57 %
Best Comparison Star	None
Optimal Aperture	4.56
Optimal Annulus	5.15
Transit Duration (day)	0.051 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.091 $\pm$ 0.039 vs 0.11 (deviation 0.49 σ)
Duration fit vs published	0.051 d vs 0.1208 d (deviation 2.49 σ)
Mid-time O-C	11.6 min
Depth SNR	2.3
Residual scatter	0.57 %

Light curve

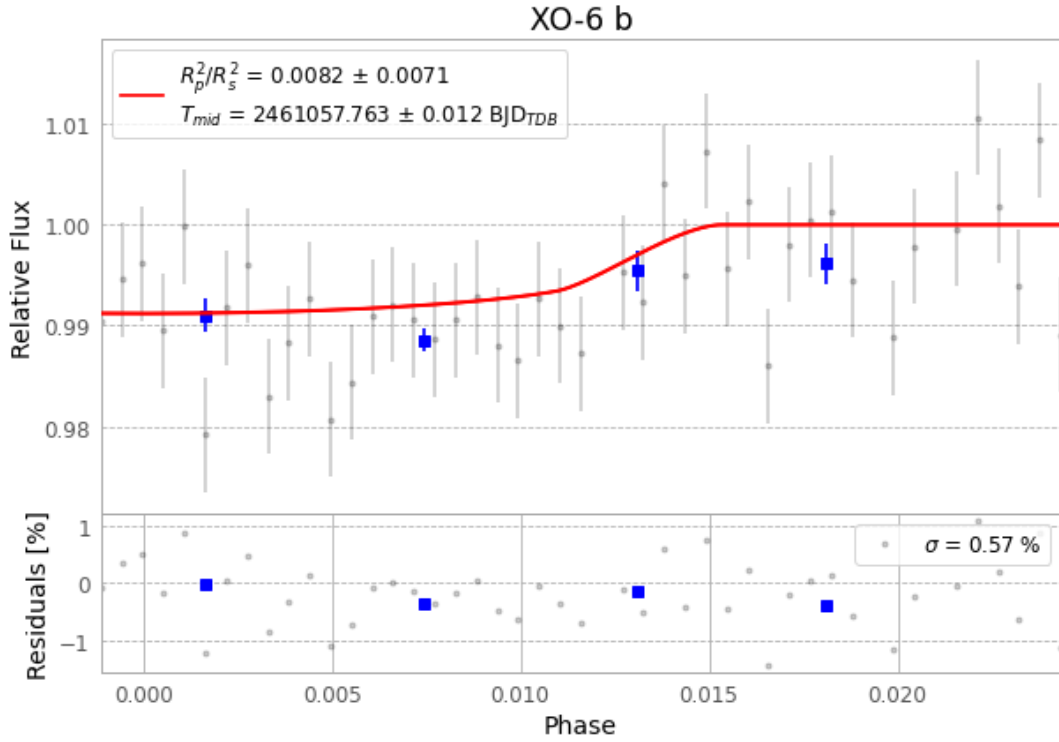


Figure 1 — FinalLightCurve_XO-6 b_2026-01-16.png (SHA-256 a08d2def56246890...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [407, 360], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 12ac924e70f7b205...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

47 FITS frames (31 MB), XO-6260117060616.FITS → XO-6260117082416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-6-260117.log (55d5f3ce0c8b...), FinalLightCurve_XO-6 b_2026-01-16.png (a08d2def5624...), FinalLightCurve_XO-6 b_2026-01-16.csv (6df1c427cac3...)

Compute resources

Wall time 121.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 00:01Z.